

rwMotorTorque

1 Executive Summary

This module maps a desired torque to control the spacecraft, and maps it to the available reaction wheels using a minimum norm inverse fit.

The optional wheel availability message is used to include or exclude particular reaction wheels from the torque solution. The desired control torque can be mapped onto particular orthogonal control axes to implement a partial solution for the overall attitude control torque.

2 Message Connection Descriptions

The following table lists all the module input and output messages. The module msg connection is set by the user from python. The msg type contains a link to the message structure definition, while the description provides information on what this message is used for.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
rwMotorTorqueOutMsg	RwMotorTorqueMsgPayload	RW motor torque output message
vehControlInMsg	CmdTorqueBodyMsgPayload	commanded vehicle control torque input message
vehControlIn2Msg	CmdTorqueBodyMsgPayload	(optional) additional commanded vehicle control torque input message
rwParamsInMsg	RWArrayConfigMsgPayload	RW array configuration input message
rwAvailInMsg	RWAvailabilityMsgPayload	(optional) RW device availability message

3 Module Parameters

The following table lists all the module parameters than can be set. The parameters are optional unless indicated (if not specified default is used).

Table 2: Module Parameters

Parameter Name	Type	Units	Default	Description	Bounds
controlAxes_B (required)	Eigen::Matrix3f	[-]	[0] _{3x3}	array of the control unit axes (axis in each row)	N/A

4 Model Assumptions and Limitations

This code makes the following assumptions:

- The number of available wheels must be equal or larger than the number of control axes. If this is not the case a zero output motor torque message is produced.

5 Initialization

The module is configured by:

```

module = rwMotorTorque.RwMotorTorque()
module.modelTag = "rwMotorTorque"
control_axes_B = [[1, 0, 0], [0, 1, 0], [0, 0, 0]]
module.controlAxes_B = control_axes_B

```

6 Detailed Model Description

6.1 Module Input and Output Behavior

This module takes an attitude control torque ${}^B\mathbf{L}_r$ and maps the vector onto the specific control axes, $\hat{\mathbf{c}}_i$. This allows only a subset of ${}^B\mathbf{L}_r$ to be implemented with the Reaction Wheels (RWs). The next step is to map reduced control torque onto the available RW spin axes, $\hat{\mathbf{g}}_{s_j}$. The module accounts for the availability of the reaction wheels in the case that not all wheels are functioning appropriately or are undergoing independent analysis.

Assume in this documentation that the number of available RWs is m , while the number of desired control axes $\hat{\mathbf{c}}_i$ is n . The number of installed RWs is N .

The commanded torque message is a required input message and is read in every time step. The RW configuration message is also required, but only read in during reset. If the RW spin axes $\hat{\mathbf{b}}_{s_i}$ change then `reset()` must be called again. The RW availability message is optional. If the availability message is not used, then all installed RWs are assumed to be available. The output message is always the array of RW motor torques.

6.2 Torque Mapping

The `rwMotorTorque` module receives a desired attitude control torque in the body frame ${}^B\mathbf{L}_r$. If two control torques are provided via the second optional control input message, these two are added together to compute ${}^B\mathbf{L}_r$. This torque is the net control torque that should be applied to the spacecraft. Let $\hat{\mathbf{g}}_{s_i}$ be the individual RW spin axis, while $\mathbf{u}_s$ is the m -dimensional array of motor torques. The $3 \times m$ projection matrix $[G_s]$ then maps the control torque on motor torques using

$$[G_s]\mathbf{u}_s = (-{}^B\mathbf{L}_r)$$

The projection matrix is defined as

$$[G_s] = [\hat{\mathbf{g}}_{s_1} \quad \cdots \quad \hat{\mathbf{g}}_{s_m}]$$

Note that here $\mathbf{u}_s$ is the array of available motor torques. The installed set of RWs could be larger than m .

The projection matrix to map a vector in the body frame $\mathcal{B}$ onto the set of control axes $\hat{\mathbf{c}}_i$ is given by

$$[CB] = \begin{bmatrix} {}^B\hat{\mathbf{c}}_1^T \\ \vdots \\ {}^B\hat{\mathbf{c}}_n^T \end{bmatrix}$$

where $n \leq 3$.

To map the requested torque onto the control axes, the first equation is pre-multiplied by $[CB]$ to yield

$$[CB][G_s]\mathbf{u}_s = [CG_s]\mathbf{u}_s = [CB](-{}^B\mathbf{L}_r) = {}^c\mathbf{L}_r$$

Note that $[CG_s]$ is a $n \times m$ matrix.

The module assumes that $m \geq n$ such that there are enough RWs available to implement ${}^c\mathbf{L}_r$. If not, then the output motor torques are set to zero with $\mathbf{u}_s = \mathbf{0}$.

To invert the equation above, a minimum norm solution is used yielding:

$$\mathbf{u}_s = [CG]^T ([CG][CG]^T)^{-1} {}^c\mathbf{L}_r$$

The final step is to map the array of available motor torques $\mathbf{u}_s$ onto the output array of installed RW motor torques.

6.3 RW Availability

If the input message `rwAvailInMsg` is linked, then the RW availability message is read in. The torque mapping is only used for RW's whose availability setting is `AVAILABLE`. If it is `UNAVAILABLE` then that RW output torque is set to zero.

7 Model Functions

The code performs the following functions:

- **Maps control torque vector onto available reaction wheels:** Takes a desired body-frame torque from `CmdTorqueBodyIntMsg` and maps it onto the RW axes.
- **Removes torque from unavailable reaction wheels:** The module observes the availability of the RWs and maps the torques to only available reaction wheels.